

1. What kind of matrix operations can be plugged in $\square$ for
10 points

$$\det(A^\square) = (\det A)^\square$$

where A is an invertible $n \times n$ matrix?

Solution. • Transpose operation:

Since a single number can be treated as a 1×1 matrix, the transpose of a scalar is itself. Then,

$$\det(A^\top) = \det(A) = (\det(A))^\top.$$

- Inverse operation:

Invertible matrix has a nonzero determinant and

$$\det(A^{-1}) = \frac{1}{\det(A)} = \det(A)^{-1}.$$

- Matrix power operator:

For any two matrix A, B where AB is defined, $\det(AB) = \det(A) \det(B)$. Then,

$$\det(A^k) = \prod_{i=1}^k \det(A) = (\det(A))^k$$

where $k \in \mathbb{N}$.

- (+3 points) if one is correct,
- (+6 points) if two are correct,
- (+10 points) if all three are correct.

2. Let a 4×4 matrix A is given as

10+10
+10+10
points

$$A = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 & 0 \end{bmatrix}.$$

- Find singular values and right-/left-singular vectors of A .
- Find $\|A\|_2$ and $\|A\|_F$.
- Find the pseudoinverse A^+ of A .
- Find a 4×4 matrix B of rank 2 that minimizes $\|A - B\|_2$.

Solution. (a) Let

$$\mathbf{a}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{a}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}.$$

Also, let

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \end{bmatrix}, \quad \text{and} \quad \mathbf{b}_3 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 0 \end{bmatrix}.$$

Then, $A = \sum_{i=1}^3 \mathbf{a}_i \mathbf{b}_i$ and $\{\mathbf{a}_i\}_{i=1}^3$, $\{\mathbf{b}_i\}_{i=1}^3$ are orthogonal sets, respectively. (+1 point)
From this, $A = \sum_{i=1}^3 \|\mathbf{a}_i\| \|\mathbf{b}_i\| \left(\frac{\mathbf{a}_i}{\|\mathbf{a}_i\|} \right) \left(\frac{\mathbf{b}_i}{\|\mathbf{b}_i\|} \right)$ where

$$\begin{aligned} \|\mathbf{a}_1\| \|\mathbf{b}_1\| &= \sqrt{2}, \\ \|\mathbf{a}_2\| \|\mathbf{b}_2\| &= \sqrt{6}, \\ \|\mathbf{a}_3\| \|\mathbf{b}_3\| &= 2\sqrt{2}. \end{aligned}$$

Let $\sigma_i = \|\mathbf{a}_{3-i}\| \|\mathbf{b}_{3-i}\|$, $\mathbf{u}_i = \frac{\mathbf{a}_{3-i}}{\|\mathbf{a}_{3-i}\|}$, and $\mathbf{v}_i = \frac{\mathbf{b}_{3-i}}{\|\mathbf{b}_{3-i}\|}$ where $i = 1, 2, 3$, then $A = \sum_{i=1}^4 \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ where $\{\mathbf{u}_i\}_{i=1}^3$ and $\{\mathbf{v}_i\}_{i=1}^3$ are orthonormal sets, respectively. So this form is the SVD expression and hence singular values are $\{2\sqrt{2}, \sqrt{6}, \sqrt{2}\}$ with $\sigma_4 = 0$, left singular vectors are

$$\left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}, \text{ and right singular vectors are } \left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

- (+1 point) if one of them is correct among 3 singular values, 3 left- and 3 right- singular vectors.
- Even if you compute full SVD for 4 singular values, left- and right- singular vectors, you get a point for each of singular vectors corresponding nonzero singular values.

- (b) From the fact that $\|A\|_2 = \max_i \sigma_i$ (+3 points) and $\|A\|_F = \sqrt{\sum_{i=1}^3 \sigma_i^2}$ (+3 points). Then, $\|A\|_2 = 2\sqrt{2}$ (+2 points) and $\|A\|_F = \sqrt{8+6+2} = 4$ (+2 points).

- If you write only the answer for each

- (c) Let $U = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$, $V = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$ and $\Sigma = \sigma_1 \mathbf{e}_1 + \sigma_2 \mathbf{e}_2 + \sigma_3 \mathbf{e}_3$ where $\{\mathbf{e}_i\}_{i=1}^3$ is the usual basis for $\mathbb{R}^3$, then $A = U\Sigma V^\top$ is the SVD of A . From this, pseudoinverse A^+ is:

$$A^+ = V\Sigma^{-1}U^\top \quad (+4 \text{ points})$$

Since the inverse matrix of diagonal matrix whose main diagonal elements are nonzero is:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{6}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \quad (+3 \text{ points})$$

Therefore,

$$\begin{aligned} A^+ &= \begin{bmatrix} 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{6}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \text{or} \\ &= \frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & 0 \end{bmatrix} + \frac{1}{6} \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \\ &\quad (+3 \text{ points}) \end{aligned}$$

- You have to use correct order in singular values and singular vectors to compute A^+ .
 - It is permitted that you did not complete computation of A^+ in one matrix.
- (d) Let $A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ be the sum of first k singular value components where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. Then, a 4×4 matrix B of rank 2 that minimizes $\|A - B\|_2$ is $B = A_2$. (+5 points)
Hence,

$$\begin{aligned} B &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 & 0 \\ 0 & \sqrt{6} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix} \quad \text{or} \\ &= 2 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & -1 \end{bmatrix} \quad (+5 \text{ points}) \end{aligned}$$

- You have to use correct order in singular values and corresponding singular vectors to compute the minima matrix B .

3. Let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ be nonzero vectors that are orthogonal to each other, that is, $\mathbf{a}_i^\top \mathbf{a}_j = 0$ for $i \neq j$. Set an $n \times n$ matrix $A = [\mathbf{a}_1 \mid \dots \mid \mathbf{a}_n]$.

10+10 points

- (a) Find singular values and right-/left-singular vectors of A .
- (b) Find $|\det A|$.

Solution. (a) First, to compute the singular values and right singular vectors,

$$A^\top A = \begin{bmatrix} \|\mathbf{a}_1\|^2 & 0 & \dots & 0 \\ 0 & \|\mathbf{a}_2\|^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \|\mathbf{a}_n\|^2 \end{bmatrix}$$

since $\{\mathbf{a}_i\}_{i=1}^n$ is orthogonal. This $A^\top A$ is diagonal, so its eigenvalues $\lambda_i(A^\top A) = \|\mathbf{a}_i\|^2$ and its corresponding eigenvectors are $\mathbf{x}_i = \mathbf{e}_i$ where $\mathbf{e}_i$ are standard basis vectors for $\mathbb{R}^n$. From the relation of $A^\top A$ and SVD, $\sqrt{\lambda_i(A^\top A)} = \|\mathbf{a}_i\|$ are singular values of A (+4 points) and $\mathbf{x}_i = \mathbf{e}_i$ are right singular vectors. (+3 points) Since each of $\mathbf{a}_i$ is nonzero vector, $\sigma_i \neq 0$. From this and relation of left- and right- singular vectors,

$$\begin{aligned} \mathbf{y}_i &= \frac{1}{\sigma_i} A \mathbf{x}_i \\ &= \frac{1}{\|\mathbf{a}_i\|} A \mathbf{e}_i = \frac{\mathbf{a}_i}{\|\mathbf{a}_i\|} \quad \text{for all } i \end{aligned} \quad (+3 \text{ points})$$

are right singular vectors.

- (b) **Method 1.** Let $U = [\mathbf{u}_1, \dots, \mathbf{u}_n]$, $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ and $\Sigma = \text{diag}(\|\mathbf{a}_1\|, \dots, \|\mathbf{a}_n\|)$, then $A = U \Sigma V^\top$. From this

$$\begin{aligned} \det(A) &= \det(U) \det(\Sigma) \det(V^\top) \\ &= \det(U) \det(\Sigma) \det(V) \end{aligned} \quad (+3 \text{ points})$$

Since U, V are orthogonal matrices in $\mathbb{R}^n$, we have $\det(U)$ and $\det(V)$ both ± 1 . (+3 points) Also, since Σ is a diagonal matrix, $\det(\Sigma) = \prod_i \|\mathbf{a}_i\|$. Therefore,

$$\begin{aligned} |\det(A)| &= |\det(U)| |\det(\Sigma)| |\det(V)| \\ &= \det(\Sigma) = \prod_{i=1}^n \|\mathbf{a}_i\|. \end{aligned} \quad (+4 \text{ points})$$

Method 2. From the properties of $\det$ and relation with eigenvalues, since $\|\mathbf{a}_i\|^2$ are the eigenvalues of $A^\top A$,

$$\det(A^\top A) = \det(A)^2 \quad (+5 \text{ points})$$

and therefore

$$\det(A) = \sqrt{\det(A^\top A)} = \sqrt{\prod_i \|\mathbf{a}_i\|^2} = \prod_i \|\mathbf{a}_i\|. \quad (+5 \text{ points})$$

4. Assume that a matrix B has eigenvalues 1, 2, 3, a matrix C has eigenvalues 4, 0, -4 , and a matrix D has eigenvalues -1 , -2 , -3 . Set a 6×6 matrix $A = \begin{bmatrix} B & C \\ \mathbf{0} & D \end{bmatrix}$ where B, C , and D are 3×3 matrices. We also define a linear transformation $\ell : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ as $\ell(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^6$.

5+5
points

- (a) What are the eigenvalues of the 6×6 matrix A ?
- (b) Consider the unit cube Q in $\mathbb{R}^6$, that is,

$$Q = \{(x_1, \dots, x_6)^\top : 0 \leq x_i \leq 1, i = 1, \dots, 6\} \subset \mathbb{R}^6.$$

Find the volume of $\ell(Q) = \{\ell(\mathbf{x}) : \mathbf{x} \in Q\}$.

Solution. (a) Since A is a block upper triangular matrix, then its set of eigenvalues is equal to the union of eigenvalues of B and D which are diagonal components in the block form. (+3 points) Therefore, the eigenvalues of A are $\{1, 2, 3, -1, -2, -3\}$. (+2 points)

- (b) The unit cube Q in $\mathbb{R}^6$ is the geometry spanned by $\{\mathbf{e}_i\}_{i=1}^6$, the standard basis vectors in $\mathbb{R}^6$. So, $\ell(Q)$ is a geometry spanned by $\{A\mathbf{e}_i\}_{i=1}^6 = \{\text{col}_i(A)\}_{i=1}^6$. (+2 points) Also, its volume is the absolute value of determinant of a matrix whose columns are elements in spanning set. Since the spanning set is the columns of A , it holds that $\text{vol}(\ell(Q)) = |\det(A)|$. Therefore, since $\det(A) = \prod_i \lambda_i(A)$,

$$\text{vol}(\ell(Q)) = \prod_{i=1}^6 \lambda_i(A) = 1^2 \times 2^2 \times 3^2 = 36. \quad (+3 \text{ points})$$

5. Suppose A is a 3×4 matrix given as
 20 points

$$A = \begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 2 & -2 & 2 \\ -2 & -1 & 1 & -1 \end{bmatrix}.$$

Find all eigenpairs of $AA^\top$.

Solution. Since $AA^\top$ is a 3×3 matrix, there are 3 eigenpairs of $AA^\top$. Notice that the SVD of A can be obtained by following:

$$\begin{aligned} A &= \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & 1 \end{bmatrix} \\ &= \sqrt{42} \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{7}} & \frac{1}{\sqrt{7}} & \frac{-1}{\sqrt{7}} & \frac{1}{\sqrt{7}} \end{bmatrix}. \end{aligned} \quad (+5 \text{ points})$$

Then, it follows that

$$\begin{aligned} AA^\top &= \sqrt{42} \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{7}} & \frac{1}{\sqrt{7}} & \frac{-1}{\sqrt{7}} & \frac{1}{\sqrt{7}} \end{bmatrix} \sqrt{42} \begin{bmatrix} \frac{2}{\sqrt{7}} \\ \frac{1}{\sqrt{7}} \\ \frac{-1}{\sqrt{7}} \\ \frac{1}{\sqrt{7}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \end{bmatrix} \\ &= 42 \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \end{bmatrix}. \end{aligned} \quad (+5 \text{ points})$$

From this, we can obtain the first eigenpair as

$$AA^\top \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} = 42 \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} = 42 \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix}.$$

In other words, $(42, \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \end{bmatrix}^\top)$ is an eigenpair of $AA^\top$. (+5 points)

Since $AA^\top$ is a rank one matrix, the remaining eigenvalues are 0 and the eigenvectors are orthogonal to $\begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \end{bmatrix}^\top$. Thus, using for example the Gram-Schmidt process, the remaining eigenvalues associated to the zero eigenvalue can be found. One example of the full eigenpairs of

$AA^\top$ are $(42, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix})$, $(0, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix})$, and $(0, \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix})$. (+5 points)

- If there are not two eigenvectors associated with 0, (−2 points).
- For each calculation mistake, (−1 points).

6. Let A be an $n \times n$ invertible matrix with an eigenpair $(\lambda, \mathbf{v})$. If $I_n + A$ is also invertible, find an eigenpair of:

points

- (a) $(I_n + A)^{-1}$;
- (b) $A + A^{-1}$;
- (c) $I_n + A^{-1}$;
- (d) $(I_n + A)^{-1} + (I_n + A^{-1})^{-1}$.

Solution. Notice that λ is neither 0 nor 1, by the assumption of problem.

(a) From the fact $(I_n + A)\mathbf{v} = (1 + \lambda)\mathbf{v}$, we can obtain $(I_n + A)^{-1}\mathbf{v} = \frac{1}{1+\lambda}\mathbf{v}$.
 $\therefore (\frac{1}{1+\lambda}, \mathbf{v})$ is an eigenpair of $(I_n + A)^{-1}$. (+5 points)

(b) $(A + A^{-1})\mathbf{v} = A\mathbf{v} + A^{-1}\mathbf{v} = \lambda\mathbf{v} + \frac{1}{\lambda}\mathbf{v} = (\lambda + \frac{1}{\lambda})\mathbf{v}$
 $\therefore (\lambda + \frac{1}{\lambda}, \mathbf{v})$ is an eigenpair of $A + A^{-1}$. (+5 points)

(c) $(I_n + A^{-1})\mathbf{v} = I_n\mathbf{v} + A^{-1}\mathbf{v} = \mathbf{v} + \frac{1}{\lambda}\mathbf{v} = (1 + \frac{1}{\lambda})\mathbf{v}$
 $\therefore (1 + \frac{1}{\lambda}, \mathbf{v})$ is an eigenpair of $I_n + A^{-1}$. (+5 points)

(d) Since $I_n + A$ is invertible, every eigenvalue of A is not -1 . It implies that every eigenvalue of A^{-1} is not -1 , so $(I_n + A^{-1})$ is invertible. From the result of (c), we can find

$$(I_n + A^{-1})^{-1}\mathbf{v} = \frac{1}{1 + \frac{1}{\lambda}}\mathbf{v} = \frac{\lambda}{1 + \lambda}\mathbf{v}.$$

Therefore,

$$\begin{aligned} ((I_n + A)^{-1} + (I_n + A^{-1})^{-1})\mathbf{v} &= (I_n + A)^{-1}\mathbf{v} + (I_n + A^{-1})^{-1}\mathbf{v} \\ &= \frac{1}{1 + \lambda}\mathbf{v} + \frac{\lambda}{1 + \lambda}\mathbf{v} \\ &= \frac{1 + \lambda}{1 + \lambda}\mathbf{v} \\ &= \mathbf{v} \end{aligned}$$

$\therefore (1, \mathbf{v})$ is an eigenpair of $(I_n + A)^{-1} + (I_n + A^{-1})^{-1}$. (+5 points)

- If you don't write the correct eigenvector, (-2 points).

7. Let A be an $m \times n$ matrix with at least one non-zero entry. What are the projection $\mathbf{P}_{\text{Col}(A)}$ onto $\text{Col}(A)$ and the pseudoinverse A^+ of A if:

points

- (a) A consists of a single column $\mathbf{v}$, that is, $n = 1$;
- (b) the columns of A are orthonormal;
- (c) the columns of A are linearly independent (that is $\text{rank } A = n$);
- (d) the columns of A are possibly linearly dependent and one of its compact SVD is given as $U\Sigma V^\top$.

Solution. Notice that $\mathbf{P}_{\text{Col}(A)} = AA^+$.

(a) We have

$$\mathbf{P}_{\text{Col}(A)} = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}, \quad A^+ = \frac{\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}. \quad (+5 \text{ points})$$

(b) Because the columns are orthonormal,

$$\mathbf{P}_{\text{Col}(A)} = AA^\top, \quad A^+ = A^\top. \quad (+5 \text{ points})$$

(c) Since $\text{rank } A = n$, $A^\top A$ is invertible.

$$\mathbf{P}_{\text{Col}(A)} = A(A^\top A)^{-1}A^\top, \quad A^+ = (A^\top A)^{-1}A^\top. \quad (+5 \text{ points})$$

(d) Let $A = U\Sigma V^\top$ be SVD of A .

$$\mathbf{P}_{\text{Col}(A)} = UU^\top, \quad A^+ = V\Sigma^{-1}U^\top. \quad (+5 \text{ points})$$

- If either $\mathbf{P}_{\text{Col}(A)}$ or A^+ is incorrect, **(−2 points)**.
- If the final answer is not in the simplest form, **(−1 points)**.

8. Show that the following matrix A is positive definite:

10 points

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \end{bmatrix}.$$

Solution. Consider a vector space $V = \{a_1 e^{-x} + a_2 e^{-2x} + a_3 e^{-3x} : a_1, a_2, a_3 \in \mathbb{R}\}$. For any $f(x), g(x) \in V$, let us define a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ by

$$\langle f, g \rangle = \int_0^\infty f(x)g(x)dx$$

Since this function is symmetric, bilinear, and positive-definite, it defines an inner product on V . (+5 points) Moreover, by simple calculation, one can verify that

$$\begin{aligned} \langle f, f \rangle &= \langle a_1 e^{-x} + a_2 e^{-2x} + a_3 e^{-3x}, a_1 e^{-x} + a_2 e^{-2x} + a_3 e^{-3x} \rangle \\ &= a_1^2 \langle e^{-x}, e^{-x} \rangle + a_2^2 \langle e^{-2x}, e^{-2x} \rangle + a_3^2 \langle e^{-3x}, e^{-3x} \rangle \\ &\quad + 2a_1 a_2 \langle e^{-x}, e^{-2x} \rangle + 2a_2 a_3 \langle e^{-2x}, e^{-3x} \rangle + 2a_3 a_1 \langle e^{-3x}, e^{-x} \rangle \\ &= \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} \langle e^{-x}, e^{-x} \rangle & \langle e^{-x}, e^{-2x} \rangle & \langle e^{-x}, e^{-3x} \rangle \\ \langle e^{-2x}, e^{-x} \rangle & \langle e^{-2x}, e^{-2x} \rangle & \langle e^{-2x}, e^{-3x} \rangle \\ \langle e^{-3x}, e^{-x} \rangle & \langle e^{-3x}, e^{-2x} \rangle & \langle e^{-3x}, e^{-3x} \rangle \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \\ &= \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \end{aligned}$$

where $f(x) = a_1 e^{-x} + a_2 e^{-2x} + a_3 e^{-3x}$. Since $\langle f, f \rangle > 0$ for every nonzero $f \in V$,

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} > 0$$

whenever $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \neq 0$. This implies the positiveness of A . (+5 points)

- Simple mistake (−1 points)
- Critical mistake (−2 points)